Software Requirements Specification (SRS)

AI Personal Health and Medicine Management System

1. Introduction

Field	Description
Purpose	To define the functional and non-functional requirements of the AI Personal Health and Medicine Management System, which enables users to manage personal health data, medicines, appointments, and medical records, supported by an AI-based health assistant.
Scope	The system provides a web-based platform for patients/users to authenticate, manage personal and medical information, track medicines with reminders, schedule appointments, maintain health records, submit feedback, and interact with an AI assistant for health queries. Admins manage users, feedback, and overall system data.
Intended Audience	Developers, project guide/evaluators, testers, and end users (patients and administrators).
SDLC Model	Scrum (iterative, sprint-based development with incremental delivery)
Technology Stack	**Frontend:** HTML, CSS, JavaScript, Bootstrap **Backend:** Python **Database:** MySQL **AI Module:** Python

2. Overall Description

Aspect	Description
Product Perspective	Standalone web application with a three-tier architecture consisting of presentation, application/logic, and data layers, along with an integrated AI module for health assistance.
User Classes	**1. User/Patient** – Registers, manages personal health data, medicines, appointments, medical records, feedback, and AI queries. **2. Admin** – Manages users, monitors/manages feedback, and oversees system data.
Operating Environment	Any modern web browser such as Chrome, Firefox, or Edge. The application runs using a Python-based backend with a MySQL database.

Bhavsar Shiv R [246040307006]
Pithadiya Pranjal N [246040307106]

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Aspect	Description
Assumptions	Users have internet access and a valid email for OTP-based verification. Medicine and appointment reminders assume that the user has enabled the required notifications.
Constraints	The system must use the specified technology stack (Python, MySQL, HTML, CSS, JavaScript, and Bootstrap). AI responses are limited to health-related queries, and the privacy and security of medical information must be maintained.

3. Functional Requirements

FR ID	Module	Requirement Description	Priority
FR-01	Authentication	System shall allow user registration and login using email/username and password, with OTP verification. System shall provide a "Forgot Password" flow with OTP-based reset.	High
FR-02	Manage Personal Information	User shall be able to add/update profile details and emergency contact information, able to search/retrieve their stored personal details.	High
FR-03	Manage Medical Report	User shall be able to upload, update, search, and delete medical reports.	High
FR-04	Manage Medicine	User shall be able to add, update, search, and delete medicine entries, generate and trigger reminders for medicine intake based on the defined schedule	High
FR-05	Manage Appointment	User shall be able to add, update, search, and delete appointment details. send reminders for upcoming appointments	High
FR-06	Manage Health Record	User shall be able to add, update, retrieve, and delete health records.	High
FR-07	Submit Feedback	User shall be able to submit feedback, which is verified and stored by the system.	Medium
FR-08	Manage Feedback	Admin shall be able to retrieve, search, and delete feedback entries.	Medium

FR ID	Module	Requirement Description	Priority
FR-09	Manage Users	Admin shall be able to add, modify, search, and remove user accounts.	High
FR-10	AI Health Assistance	System shall accept user health queries and return AI-generated chat responses. generate AI-based health recommendations and summaries and can export it as pdf , image etc at the time of conversation.	High

4. Non-Functional Requirements

NFR ID	Category	Requirement Description
NFR-01	Performance	System shall respond to standard user requests within 2–3 seconds under normal load.
NFR-02	Security	Passwords shall be stored using secure hashing, and OTP shall be used for sensitive actions such as authentication and password reset. Medical data shall be accessible only to the authenticated owner and authorized administrators.
NFR-03	Usability	User interface shall be responsive and usable across desktop and mobile browsers.
NFR-04	Reliability	System shall maintain data consistency and provide acknowledgement/confirmation for create, update, and delete operations.
NFR-05	Scalability	Database and application design shall support increasing numbers of users and records without major redesign.
NFR-06	Maintainability	Code shall follow a modular and organized structure to support easy maintenance, testing, and Scrum-based iterative development.
NFR-07	Availability	System shall aim for high availability to support medicine reminders and appointment notification features.

5. External Interface Requirements

Interface Type	Description
User Interface	Web pages developed using HTML, CSS, JavaScript, and Bootstrap for registration, login, user and admin dashboards, medicine and appointment forms, health records, medical reports, and AI chat interaction.

Interface Type	Description
Hardware Interface	Standard client devices such as PCs, laptops, and smartphones with internet connectivity. Server hardware shall be capable of running the Python-based application and MySQL database.
Software Interface	Python for backend application logic, MySQL for database management, Python-based AI libraries for natural language and health query processing, and an email/OTP service for verification.
Communication Interface	HTTPS shall be used for secure communication between the client and server. The frontend shall communicate with the Python backend through appropriate application requests and responses.

6. System Actors and Use Cases

Actor	Use Cases
User/Patient	Authentication, Manage Personal Information, Manage Medical Report, Manage Medicine, Manage Appointment, Manage Health Record, Submit Feedback, AI Health Assistance
Admin	Authentication, Manage Users, Manage Medical Report, Manage Medicine, Manage Appointment, Manage Health Record, Manage Feedback

7. Future Enhancements

Enhancement	Description
Mobile App	Development of a native mobile application with push-based medicine and appointment reminders.
Wearable Integration	Integration with fitness and health wearables for collecting and synchronizing health-related information.
Advanced AI	Implementation of advanced AI capabilities for predictive health analysis based on historical health records.

8. System Models

- Context Diagram
- DFD Level 1 ,2
- Use Case Diagram
- ER Diagram